

Vayu: Verification and Test Evidence

Supporting document, SIH 2026, ByteBrigade. States what has been tested and what has not.

1. Automated tests

95 unit tests, all passing (Python unittest, run from the etl-python folder). They test code behaviour on fixtures. They do not validate the index against official statistics.

Test module	Tests	What it covers
test_normalize	3	Grouping by route, median price, multiple sources, empty input
test_anomaly_filter	4	Day-over-day spike dropped, first observation passes, cross-sectional outlier dropped, skip when too few routes
test_index_engine	4	Weighted average at full coverage, renormalisation at partial coverage, fallback display price, zero-coverage behaviour
test_models	7	Fare record schema and the Rs 1,000 to Rs 1,00,000 sanity band
test_xhr_parser	10	Generic response parser, including planted mislabelled fields that must be rejected
test_fare_scraper_xhr_parsing	17	Scraper handling of parsed responses
test_google_flights_batchexecute	9	Parser for Google Flights' response format, against a real captured response
test_indigo_fare_radar	7	IndiGo fare-radar parser, against a real captured response
test_airindia_air_bounds	8	Air India parser, against a real captured response (multiple fare families per flight, airport-to-city code mapping)
test_google_flights_page_action	4	Consent-screen handling for the browser automation
test_air_india_page_action	22	Form-filling automation and its failure paths

2. Real-data evidence

Three test fixtures are real responses captured from live sites on 2026-09-27 (Google Flights, IndiGo, Air India), not invented data. Tests built on them caught real defects: a file-encoding bug on Windows that corrupted the rupee symbol, a parser that returned zero fares from a valid response, and a phantom route created by a second Mumbai airport code.

3. Live pipeline runs

The scheduled workflow ran the full 325-route basket against Google Flights. A recorded full run took about 88 minutes, and the log showed successful responses with no errors. The latest dashboard reading computed the index from 212 of 325 routes (85.9% of basket weight). Run logs and screenshots are in the evidence folder.

4. Not verified

Item	Status
Index level against official statistics	Not validated. The reference prices are modelled, so the level is not comparable to CPI-style series (see the methodology note).
Basket weights	Illustrative gravity model, not DGCA traffic data.
IndiGo and Air India in the scheduled run	Parsers tested on real captured responses, but these sources are not part of the daily workflow.
MakeMyTrip	Tested and blocked by bot defences.
SpiceJet, Akasa	Not implemented.
Behaviour on all 325 routes	Coverage varies by run. Some routes return no fares on a given day and are excluded from that day's index.
Dashboard build and types	Reviewed and deployed. No automated front-end tests.